

Complete HTML Guide

This covers all the topics we have included in Module 01 (HTML) for Month 01

3. Basic HTML Structure

DOCTYPE Declaration

The `<!DOCTYPE html>` declaration defines the document type and version of HTML. It must be the first line in your HTML document.

Essential HTML Tags

- `<html>` - Root element that wraps all content on the page
- `<head>` - Contains metadata about the document (not displayed on page)
- `<body>` - Contains all visible content of the webpage

Adding Titles and Metadata

- `<title>` - Sets the page title shown in browser tabs and search results
- `<meta>` - Provides metadata like character encoding, viewport settings, and descriptions
- `<link>` - Links external resources like stylesheets
- `<script>` - Includes JavaScript code or external scripts

Basic HTML Template

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Page Title</title>
</head>
<body>
  <!-- Content goes here -->
</body>
</html>
```

4. Basic HTML Tags

Heading Tags

HTML provides six levels of headings, from most important to least important:

- **<h1>** - Main heading (largest)
- **<h2>** - Secondary heading
- **<h3>** - Third-level heading
- **<h4>** - Fourth-level heading
- **<h5>** - Fifth-level heading
- **<h6>** - Sixth-level heading (smallest)

Paragraph Tag

- **<p>** - Defines a paragraph of text with automatic spacing before and after

Anchor Tag

- **<a>** - Creates hyperlinks to other pages, files, or sections
- **Attributes:**
 - **href** - Specifies the URL or destination
 - **target** - Defines where to open the link (e.g., **_blank** for new tab)

Image Tag

- **** - Embeds images in the webpage (self-closing tag)
- **Attributes:**
 - **src** - Specifies the image file path or URL
 - **alt** - Provides alternative text for accessibility and SEO

Text Formatting Tags

- **** - Makes text bold (visual emphasis)
- **<i>** - Makes text italic (visual emphasis)
- **<u>** - Underlines text
- **<big>** - Makes text larger (deprecated in HTML5)
- **<small>** - Makes text smaller

Line Breaks and Rules

- **
** - Inserts a line break (self-closing tag)
- **<hr>** - Creates a horizontal rule/line (self-closing tag)

Subscript and Superscript

- `<sub>` - Creates subscript text (below baseline)
- `<sup>` - Creates superscript text (above baseline)

Preformatted Text

- `<pre>` - Preserves whitespace and line breaks exactly as written in the code

5. Page Layout and Semantics

Semantic HTML5 Tags

These tags provide meaning and structure to your content:

- `<header>` - Contains introductory content or navigation
- `<main>` - Represents the main content of the document
- `<footer>` - Contains footer information like copyright or contact details
- `<section>` - Defines a distinct section of content
- `<article>` - Represents standalone, self-contained content
- `<aside>` - Contains content related to the main content (sidebars, etc.)
- `<nav>` - Contains navigation links

Generic Container Tags

- `<div>` - Generic block-level container for grouping elements
- `<span>` - Generic inline container for styling specific parts of text

SEO-Friendly Page Structure

A well-structured page improves search engine optimization:

- Use semantic tags appropriately
- Include proper heading hierarchy (h1, h2, h3, etc.)
- Add meta descriptions and titles
- Use alt attributes for images
- Structure content logically

6. Lists, Tables, and Forms

Unordered Lists

- **** - Creates a bulleted list
- **** - Defines each list item

Ordered Lists

- **** - Creates a numbered list
- **** - Defines each list item (same as unordered lists)

Tables

- **<table>** - Creates a table
- **<tr>** - Defines a table row
- **<td>** - Defines a table cell (data)
- **<th>** - Defines a table header cell
- **<caption>** - Provides a table caption/title
- **<thead>** - Groups header content
- **<tbody>** - Groups body content

Table Attributes

- **colspan** - Makes a cell span multiple columns
- **rowspan** - Makes a cell span multiple rows

Forms

- **<form>** - Creates a form container
- **<input>** - Creates various input fields (text, email, password, etc.)
- **<textarea>** - Creates a multi-line text input
- **<select>** - Creates a dropdown menu
- **<button>** - Creates a clickable button

Common Input Types

- **type="text"** - Single-line text input
- **type="email"** - Email address input
- **type="password"** - Password input (hidden text)
- **type="submit"** - Submit button
- **type="checkbox"** - Checkbox input
- **type="radio"** - Radio button input

7. Multimedia Embedding

Video Embedding

- **<video>** - Embeds video content
- **Attributes:**
 - **src** - Specifies the video file path or URL
 - **controls** - Adds video controls (play, pause, volume)
 - **autoplay** - Automatically starts playing (use carefully)
 - **loop** - Repeats the video continuously
 - **width** and **height** - Set video dimensions
 - **poster** - Shows an image before video plays

Video Example

```
<video controls width="640" height="360">  
  <source src="video.mp4" type="video/mp4">  
  <source src="video.webm" type="video/webm">  
  Your browser does not support the video tag.  
</video>
```

Embedding YouTube Videos

Use the iframe method to embed YouTube videos:

```
<iframe width="560" height="315"  
  src="https://www.youtube.com/embed/VIDEO_ID"  
  frameborder="0"  
  allowfullscreen>  
</iframe>
```

Audio Embedding

- **<audio>** - Embeds audio content
- **Attributes:**
 - **src** - Specifies the audio file path or URL
 - **controls** - Adds audio controls
 - **autoplay** - Automatically starts playing
 - **loop** - Repeats the audio continuously

Additional Multimedia Tags

- **<iframe>** - Embeds external content (videos, maps, other websites)
 - **<embed>** - Embeds external content or plugins
 - **<object>** - Embeds multimedia content or external resources
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